



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2014

YEAR 11

ASSESSMENT TASK #1

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Question is to be returned in a separate bundle.
- All necessary working should be shown in every question.
- Answers should be in simplest exact form unless specified otherwise.

Total Marks – 60

- Attempt questions 1 – 4.
- Each question is worth 15 marks.

Examiner: *E. Choy & A. Fuller*

Question One (15 marks)

- (a) By considering the expansion of $\cos(\alpha - \beta)$, find the exact value of $\cos 15^\circ$. [2]
- (b) Find the coordinates of the point P which divides the interval from $A(-1, 5)$ to $B(6, -4)$ externally in the ratio $2 : 3$. [2]
- (c) The acute angle between the lines $x + y = 3$ and $3x - y = 1$ is θ . [2]
What is the value of $\tan \theta$?
- (d) If α, β, γ are the roots of $2x^3 - 4x^2 - 3x - 1 = 0$, find the value [7]
of the following:
- (i) $\alpha + \beta + \gamma$
 - (ii) $\alpha\beta + \alpha\gamma + \beta\gamma$
 - (iii) $\alpha\beta\gamma$
 - (iv) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$
 - (v) $(\alpha - 1)(\beta - 1)(\gamma - 1)$
 - (vi) $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$
- (e) Using the t -results, or otherwise, prove that [2]
 $\cot \frac{\theta}{2} - \cot \theta = \operatorname{cosec} \theta$.

Question Two (15 marks) **(Use a SEPARATE writing booklet)**

- (a) (i) Show that $(x + 2)$ is a factor of $P(x) = x^3 - 2x^2 - 20x - 24$. [3]
- (ii) Hence, fully factorise $P(x) = x^3 - 2x^2 - 20x - 24$.
- (b) Find the general solution of $2 \cos 2x = 1$. [2]
- (c) Find all values of x such that $x + \frac{36}{x} \geq 13$. [3]
- (d) By substituting $y = \sin x + \cos x$ into $\sin 2x + 1 = 5 \sin x + 5 \cos x$, [3]
- (i) show that $y^2 - 5y = 0$.
- (ii) Hence, or otherwise, find all solutions between 0 and 2π of
- $\sin 2x + 1 = 5 \sin x + 5 \cos x$.
- (e) (i) By repeated substitution of $x = 1$, and differentiation, [4]
- or otherwise, find constants a, b, c, d and e such that
- $x^4 - 4x^3 + x^2 + 6x + 2 \equiv a(x - 1)^4 + b(x - 1)^3 + c(x - 1)^2 + d(x - 1) + e$
- (ii) Hence, or otherwise, solve $x^4 - 4x^3 + x^2 + 6x + 2 = 0$

Question Three (15 marks) (Use a SEPARATE writing booklet)

- (a) The point $A(a, c)$ lies on the line $4x + 3y - 30 = 0$ and [6]

the point $B(b, d)$ lies on the line $x + 3y + 15 = 0$.

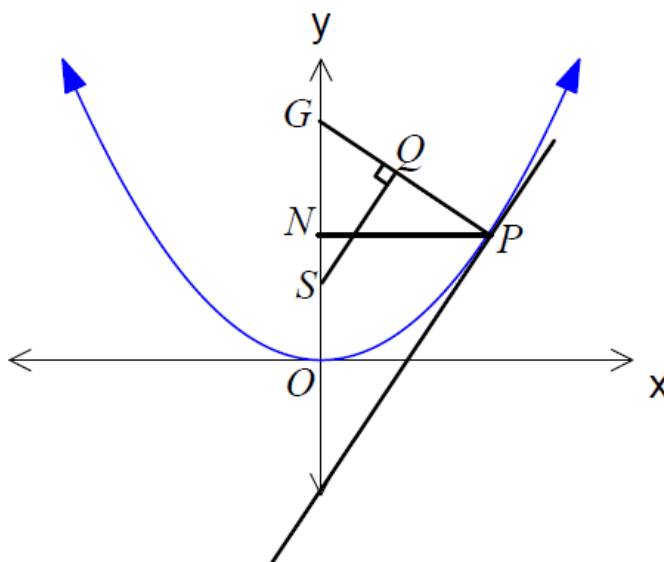
If $P(3, 2)$ divides the interval AB in the ratio $2 : 1$.

- (i) Show that $a = 9 - 2b$ and $c = 6 - 2d$.

- (ii) Explain why $b + 3d + 15 = 0$.

- (iii) Find the coordinates of A and B .

- (b) Consider the variable point $P(2at, at^2)$ on the parabola $x^2 = 4ay$. [9]



- (i) Show that the equation of the normal at P is $x + ty = at^3 + 2at$
- (ii) The line from S , the focus of the parabola, meets the normal at P at right angles at Q , as shown.
- Show that Q has coordinates $(at, a(t^2 + 1))$.
- (iii) Show that the locus of Q is a parabola and state its focal length.
- (iv) The line from P , parallel to the x -axis, meets the y -axis at N .

By finding the perpendicular distance of the focus to the normal, or otherwise, show that $SQ^2 = ON \times SP$.

Question Four (15 marks) **(Use a SEPARATE writing booklet)**

- (a) Without the use of calculus, find the range of the following: [5]

(i) $y = \sin^2 x - 2 \sin x + 2$

(ii) $y = \cos x - 2 \sin x + 2$

- (b) Given a parabola $P : y^2 = 4x$. [10]

Let the line $L : y = mx + c$ be a tangent to P .

(i) Show that $c = \frac{1}{m}$.

- (ii) Suppose that L passes through the point (h, k) . Using the above result, show that $hm^2 - km + 1 = 0$.

- (iii) If the slopes of the two tangents from the point (h, k) to the parabola P are m_1 and m_2 , show that $(m_1 - m_2)^2 = \frac{k^2 - 4h}{h^2}$.

- (iv) Q is a point on the circle $x^2 + y^2 = 20$. The slopes of the two tangents from Q to P are s_1 and s_2 such that $\left(\frac{1}{s_2} - \frac{1}{s_1}\right)^2 = 8$.

Find the coordinates of Q .

- (v) R is a point such that the angle between the two tangents from R to P is 45° . Find the equation of the locus of R .

End of Examination

2014 Mathematics Extension Task 1: Solutions

Question 1 (15 marks)

- (a) By considering the expansion of $\cos(\alpha - \beta)$, find the exact value of $\cos 15^\circ$.

2

Solution: $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$.

$$\begin{aligned}\cos 15^\circ &= \cos(45^\circ - 30^\circ) \text{ [or } \cos(60^\circ - 45^\circ)], \\ &= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \text{ [or } \cos 60^\circ \cos 45^\circ + \sin 60^\circ \sin 45^\circ], \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \left[\text{or } \frac{1}{2} \cdot \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} \right], \\ &= \frac{\sqrt{3} + 1}{2\sqrt{2}} \text{ or } \frac{\sqrt{2}(\sqrt{3} + 1)}{4}.\end{aligned}$$

- (b) Find the coordinates of the point P which divides the interval from $A(-1, 5)$ to $B(6, -4)$ externally in the ratio $2 : 3$.

2

Solution:
$$x_P = \frac{3(-1) - 2(6)}{3 - 2} = -15, \quad y_P = \frac{3(5) - 2(-4)}{3 - 2} = 23.$$

So P is $(-15, 23)$.

- (c) The acute angle between the lines $x + y = 3$ and $3x - y = 1$ is θ . What is the value of $\tan \theta$?

2

Solution: $y = -x + 3$, so $m_1 = -1$,
 $y = 3x - 1$, so $m_2 = 3$,
 $\therefore \tan \theta = \frac{|-1 - 3|}{1 + (-1)3},$
 $= 2.$

- (d) If α, β, γ are the roots of $2x^3 - 4x^2 - 3x - 1 = 0$, find the value of the following:
 (i) $\alpha + \beta + \gamma$

7

Solution:
$$\alpha + \beta + \gamma = -\left(\frac{-4}{2}\right),$$

 $= 2.$

- (ii) $\alpha\beta + \alpha\gamma + \beta\gamma$

Solution:
$$\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{3}{2}.$$

(iii) $\alpha\beta\gamma$

Solution:
$$\alpha\beta\gamma = -\left(\frac{-1}{2}\right),$$
$$= \frac{1}{2}.$$

(iv) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

Solution:
$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma},$$
$$= -\frac{3}{2} \div \frac{1}{2},$$
$$= -3.$$

(v) $(\alpha - 1)(\beta - 1)(\gamma - 1)$

Solution:
$$(\alpha - 1)(\beta - 1)(\gamma - 1) = \alpha\beta\gamma - (\alpha\beta + \alpha\gamma + \beta\gamma) + (\alpha + \beta + \gamma) - 1,$$
$$= \frac{1}{2} + \frac{3}{2} + 2 - 1,$$
$$= 3.$$

(vi) $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$

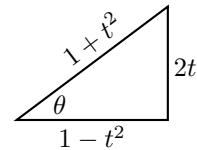
Solution:
$$(\alpha\beta + \alpha\gamma + \beta\gamma)^2 = \alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 + 2(\alpha\beta^2\gamma + \alpha^2\beta\gamma + \alpha\beta\gamma^2),$$
$$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma),$$
$$= \frac{9}{4} - 2 \cdot \frac{1}{2} \cdot 2,$$
$$= \frac{1}{4}.$$

(e) Using the t -results, or otherwise, prove that $\cot \frac{\theta}{2} - \cot \theta = \operatorname{cosec} \theta$.

2

Solution: Put $\tan \frac{\theta}{2} = t$.

L.H.S.
$$= \frac{1}{t} - \frac{1 - t^2}{2t},$$
$$= \frac{2 - 1 + t^2}{2t},$$
$$= \frac{1 + t^2}{2t},$$
$$= \frac{1}{\sin \theta},$$
$$= \text{R.H.S.}$$



QUESTION 2.

$$(a) (i) \quad P(-2) = -8 - 8 + 40 - 24 \\ = 0.$$

[1]

$\therefore x+2$ is a factor.

$$(ii) \quad \begin{array}{r} x+2 \overline{) x^3 - 2x^2 - 20x - 24} \\ \underline{x^3 + 2x^2} \\ -4x^2 - 20x \\ \underline{-4x^2 + 8x} \\ -12x - 24 \\ \underline{-12x - 24} \\ 0 \end{array}$$

$$\begin{aligned} \therefore P(x) &= (x+2)(x^2 - 4x - 12) \\ &= (x+2)(x-6)(x+2) \\ &= (x+2)^2(x-6) \end{aligned}$$

[2]

OR.

$$\begin{array}{r|rrrr} & 1 & -2 & -20 & -24 \\ -2 & & -2 & 8 & 24 \\ \hline & 1 & -4 & -12 & 0 \end{array}$$

$$\therefore (x+2)(x^2 - 4x - 12) = (x+2)^2(x-6)$$

OR. $x^3 - 2x^2 - 20x - 24 = (x+2)(x^2 + bx + c)$

$$\begin{aligned} \text{Clearly } 2c &= -24 \\ c &= -12. \end{aligned}$$

$$\text{4 w-coeff of } x \text{ is } 2b + c = -20$$

$$2b - 12 = -20$$

$$2b = -8$$

$$b = -4$$

$$\therefore (x+2)(x^2 - 4x - 12) = (x+2)^2(x-6).$$

$$(b) \quad 2 \cos x = 1.$$

$$\cos x = \frac{1}{2}$$

$$2x = 2n\pi \pm \frac{\pi}{3} \quad ; \quad n \in \mathbb{Z}.$$

$$x = n\pi \pm \frac{\pi}{6} \quad ; \quad n \in \mathbb{Z}.$$

[2]

$$(c) \quad x + \frac{36}{x} \geq 13.$$

$$\therefore x^3 + 36x \geq 13x^2$$

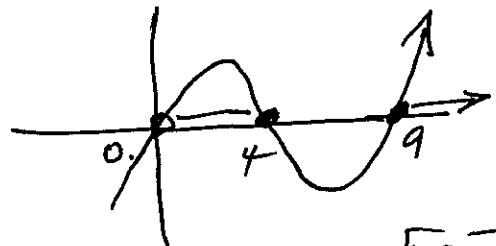
($\times$ both sides by x^2)

$$x^3 - 13x^2 + 36x \geq 0.$$

$$x(x^2 - 13x + 36) \geq 0$$

$$x(x-4)(x-9) \geq 0.$$

$$\therefore \boxed{0 \leq x \leq 4, \quad x \geq 9}$$



[3]

$$(d) (i) \sin 2x + 1 = 2 \sin x \cos x + \sin^2 x + \cos^2 x \\ = (\sin x + \cos x)^2.$$

$$\therefore y^2 = 5y. \\ \text{OR } y^2 - 5y = 0.$$

[2]

$$(ii) \therefore \text{if } y^2 - 5y = 0 \\ y = 0 \text{ or } 5$$

$$\text{i.e. } \sin x + \cos x = 0 \quad \text{OR} \quad \sin x + \cos x = 5$$

$$\sin x = -\cos x$$

NO. SOLNS!!

$$\frac{\sin x}{\cos x} = -1$$

$$\tan x = -1$$

$$\tan x = -1$$

$$\boxed{x = \frac{3\pi}{4}, \frac{7\pi}{4}}$$

[1]

$$(e) \text{ Now } x^4 - 4x^3 + x^2 + bx + 2 \equiv a(x-1)^4 + b(x-1)^3 + c(x-1)^2 + d(x-1) + e \quad (A)$$

Clearly Monic $\therefore \boxed{a=1}$

$$\text{Also if } \underline{x=1}$$

$$1 - 4 + 1 + b + 2 = e$$

$$\boxed{e=b}$$

Differentiate both sides of (A) where $a=1$

$$4x^3 - 12x^2 + 2x + b \equiv 4(x-1)^3 + 3b(x-1)^2 + 2d(x-1) + d \quad (B)$$

$$\text{Let } \underline{x=1}$$

(e) (CONTD)

$$4 - 12 + 2 + b = d.$$

$$d = 0.$$

Differentiate (B)

$$12x^2 - 24x + 2 = 12(x-1)^2 + 6b(x-1) + 2c. \quad (C)$$

$$\text{let } x = 1$$

$$12 - 24 + 2 = 2c$$

$$\boxed{c = -5}$$

Differentiate (C)

$$24x - 24 = 24(x-1) + 6b$$

$$\text{let } x = 1$$

$$0 = 6b$$

$$\boxed{b = 0.}$$

$$\therefore \boxed{x^4 - 4x^3 + x^2 + 6x + 2 \equiv (x-1)^4 - 5(x-1)^2 + 6.}$$

(11.) Hence $(x-1)^4 - 5(x-1)^2 + 6 = 0$

$$\text{let } (x-1)^2 = u$$

$$u^2 - 5u + 6 = 0$$

$$(u-3)(u-2) = 0$$

$$u = 2, 3$$

$$\therefore (x-1)^2 = 2$$
$$\boxed{x = 1 \pm \sqrt{2}}$$

$$\text{OR } (x-1)^2 = 3$$
$$\boxed{x = 1 \pm \sqrt{3}}$$

HSC Maths Ext 1 Assessment Task 1 Question 3

$$a). (i) (3, 2) = \left(\frac{2b+a}{3}, \frac{2d+c}{3} \right)$$

$$\frac{2b+a}{3} = 3 \quad \left(\frac{1}{2} \right)$$

$$\frac{2d+c}{3} = 2 \quad \left(\frac{1}{2} \right)$$

$$2b+a=9$$

$$\boxed{a=9-2b} \quad \left(\frac{1}{2} \right)$$

$$2d+c=6$$

$$\boxed{c=6-2d} \quad \left(\frac{1}{2} \right)$$

(ii) $b+3d+15=0$ because $B(b, d)$ lies on the line $x+3y+15=0$ (1)

$$(iii) a=9-2b \text{ — (1)}$$

$$c=6-2d \text{ — (2)}$$

$$4a+3c-30=0 \text{ — (3)}$$

$$b+3d+15=0 \text{ — (4)}$$

sub (1) & (2) in (3),

$$4(9-2b) + 3(6-2d) - 30 = 0$$

$$36 - 8b + 18 - 6d - 30 = 0$$

$$8b + 6d = 24$$

$$4b + 3d = 12 \text{ — (5)} \quad (1)$$

sub (4) in (5),

$$4(-3d-15) + 3d = 12$$

$$-12d - 60 + 3d = 12$$

$$-9d = 72$$

$$\boxed{d = -8} \quad \left(\frac{1}{2} \right)$$

$$\boxed{b = 9} \quad \left(\frac{1}{2} \right)$$

$$\boxed{a = -9} \quad \left(\frac{1}{2} \right)$$

$$\boxed{c = 22} \quad \left(\frac{1}{2} \right)$$

b). (i) $x^2 = 4ay$

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

At P, $m_T = \frac{2at}{2a} = t$

$$m_N = -\frac{1}{t} \quad (1)$$

The eqⁿ of normal at P,

$$y - at^2 = -\frac{1}{t}(x - 2at)$$

$$ty - at^3 = -x + 2at$$

$$x + ty = at^3 + 2at \quad (1)$$

(1)

(ii) The eqⁿ of SQ,

$$m_{SQ} = t, \quad S(0, a)$$

$$y - a = t(x - 0)$$

$$y = tx + a$$

(2)

Sub (2) in (1),

$$x + t(tx + a) = at^3 + 2at$$

$$x + t^2x + at = at^3 + 2at$$

$$(1 + t^2)x = at^3 + at$$

$$(1 + t^2)x = at(t^2 + 1) \quad \text{since } t^2 + 1 \neq 0$$

$$\boxed{x = at} \quad (1)$$

$$y = at(t) + a$$

$$y = at^2 + a$$

$$\boxed{y = a(t^2 + 1)}$$

$$\therefore Q(at, a(t^2 + 1)) \quad (1)$$

(1)

b) (iii) $x = at$, $y = a(t^2 + 1)$

$$y = a\left(\frac{x^2}{a^2} + 1\right)$$

$$ay = x^2 + a^2 \quad (1)$$

$x^2 = a(y - a)$ which is a parabola with focal length $\frac{a}{4}$. (1)

(iv) $N(0, at^2)$
 $S(0, a)$
 $Q(at, a(t^2 + 1))$
 $P(2at, at^2)$

$$ON = at^2 \quad \left(\frac{1}{2}\right)$$

$$\begin{aligned} SP &= PM \quad (\text{By definition}) \\ &= at^2 + a \\ &= a(t^2 + 1) \quad \left(\frac{1}{2}\right) \end{aligned}$$

$$\begin{aligned} SQ &= \sqrt{(at - 0)^2 + (at^2 + a - a)^2} \\ &= \sqrt{a^2 t^2 + a^2 t^4} \\ &= \sqrt{a^2 t^2 (1 + t^2)} \\ &= at \sqrt{t^2 + 1} \end{aligned}$$

$$SQ^2 = a^2 t^2 (t^2 + 1) \quad (1)$$

$$ON \times SP = at^2 \times a(t^2 + 1)$$

$$= a^2 t^2 (t^2 + 1)$$

$$= SQ^2 \quad (1)$$

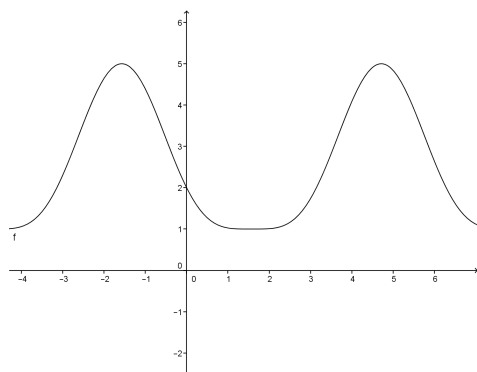
Question 4

(a) (i) $y = \sin^2 x - 2 \sin x + 2$
 $= (\sin x - 1)^2 + 1$

Considering min and max values of $(\sin x - 1)^2$,

Range: $1 \leq y \leq 5$

[3]

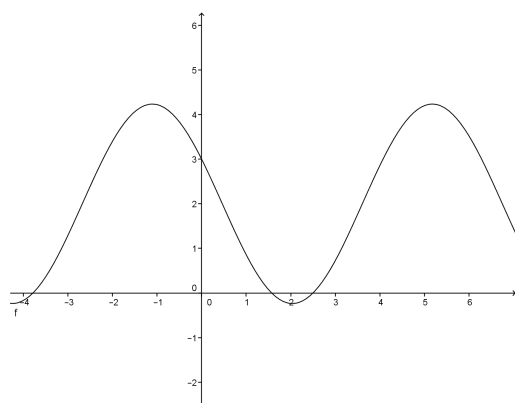


(ii) $y = \cos x - 2 \sin x + 2$
 $= \sqrt{5} (\cos(x + \alpha)) + 2$

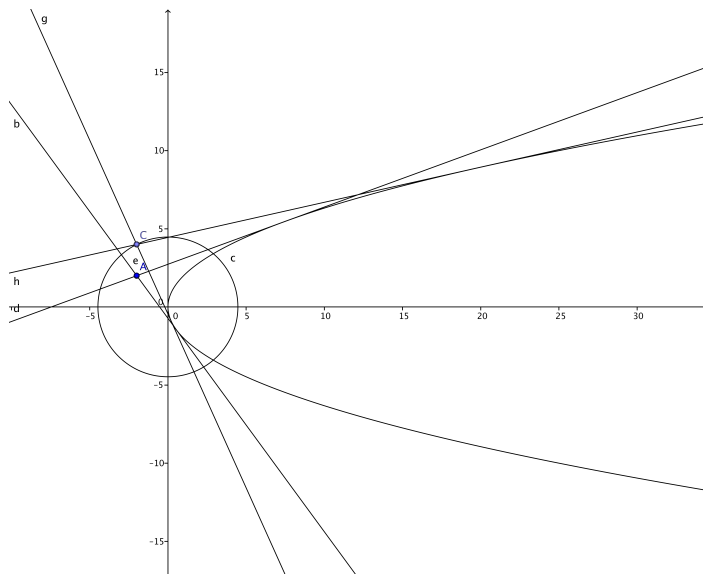
Considering max and min values of cosine,

Range: $2 - \sqrt{5} \leq y \leq 2 + \sqrt{5}$

[2]



(b)



(i) Curve: $y^2 = 4x$ Line: $y = mx + c$

Tangent: $(mx + c)^2 = 4x$
 $m^2x^2 + 2mcx + c^2 - 4x = 0$
 $m^2x^2 + (2mc - 4)x + c^2 = 0$

For the line to be a tangent, discriminant is zero.

$$\begin{aligned}(2mc - 4)^2 - 4m^2c^2 &= 0 \\ [(2mc - 4) - 2mc][(2mc - 4) + 2mc] &= 0 \\ -4[(2mc - 4) + 2mc] &= 0 \\ 4mc - 4 &= 0 \\ mc &= 1 \\ \therefore c &= \frac{1}{m} \quad [2]\end{aligned}$$

(ii) $y = mx + c$

$$\begin{aligned}k &= mh + \frac{1}{m} \\ mk &= m^2h + 1 \\ \therefore hm^2 - mk + 1 &= 0 \quad [2]\end{aligned}$$

(iii) Solutions using quadratic formula

$$m = \frac{-k \pm \sqrt{k^2 - 4h}}{2h}$$

$$\begin{aligned}
\text{Thus } (m_1 - m_2) &= \frac{-k + \sqrt{k^2 - 4h}}{2h} - \frac{-k - \sqrt{k^2 - 4h}}{2h} \\
&= \frac{2\sqrt{k^2 - 4h}}{2h} \\
(m_1 - m_2)^2 &= \frac{k^2 - 4h}{h^2} \quad [2]
\end{aligned}$$

$$(iv) \quad \left(\frac{1}{s_1} - \frac{1}{s_2} \right)^2 = \left(\frac{s_2 - s_1}{s_1 s_2} \right)^2 = 8$$

$$\text{From } hm^2 - mk + 1 = 0, \quad \alpha\beta = \frac{c}{a} = \frac{1}{h}$$

$$\begin{aligned}
\frac{(s_1 - s_2)^2}{(s_1 s_2)^2} &= \frac{k^2 - 4h}{h^2} \\
&= \frac{\left(\frac{1}{h^2} \right)^2}{1} \\
&= k^2 - 4h
\end{aligned}$$

$$\therefore k^2 - 4h = 8$$

But Q is (h, k) , and $h^2 + k^2 = 20$, so

$$h^2 + (8 + 4h) = 20, \text{ and hence}$$

$$h^2 + 4h - 12 = 0$$

$$\therefore h = -6 \text{ or } 2$$

On substitution in the equation of the circle, -6 is extraneous.
Thus $h = 2, k = \pm 4$.

$$\text{Hence } Q \text{ is } (2, 4) \text{ or } (2, -4). \quad [2]$$

(v) If θ is the angle between two lines, then

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\text{Here } \tan \theta = \tan 45^\circ = 1$$

$$\text{Hence } 1 = \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right)^2$$

$$\begin{aligned}
 &= \frac{k^2 - 4h}{h^2} \div \left(1 + 2m_1m_2 + (m_1m_2)^2\right) \\
 &= \frac{k^2 - 4h}{h^2} \times \frac{1}{1 + 2\left(\frac{1}{h}\right) + \frac{1}{h^2}} \\
 &= \frac{k^2 - 4h}{h^2 + 2h + 1}
 \end{aligned}$$

$$\therefore h^2 + 2h + 1 = k^2 - 4h$$

$$\text{ie } (h+3)^2 - k^2 = 8$$

On renaming the variables, the locus is

$$(x+3)^2 - y^2 = 8 \quad (\text{an Hyperbola, conjugate axis } x=-3) \quad [2]$$

